

Fenestrations

Tiny Retreat Research Report

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1. Introduction

Windows, skylights and doors are fenestration systems that are essential to managing lighting, ventilation, and thermal performance. For a tiny retreat with small floor areas and a high ratio of exterior surface to interior space, fenestration design is optimal to promote consistent comfort and energy efficiency. Well-placed windows and openings can enhance daylighting and natural ventilation, reducing the need for mechanical HVAC systems and maximizing the use of passive resources to save energy.

This tiny retreat can present several issues, such as a lack of exposure to greenery, insufficient lighting, high noise levels, and more. Due to these factors, choosing the right fenestration is crucial and should solve these problems as effectively as possible. This analysis focuses on two operable window types, casement windows and awning windows, which use crank operation and support insect screens and determines which is most suitable for this retreat.

2. Analysis of Window Options

2.1 Casement Windows

Casement windows are side-hinged and open outward by crank operation. These units open outward like a door and provide excellent ventilation, unobstructed views and are really well known for their large usability in different architectural contexts (*Casement Windows 101 (Everything You NEED to Know)*, 2024). In addition, having the hinge at the side and open outwards fully can provide maximum ventilation and natural lighting to the retreat, which contributes a lot to one's comfort level during the summer (*Awning vs. Casement Windows*, n.d.). During the winter, as the temperature drops, the casement window's tight sealing performance when closed will keep the room insulated and prevent the cold air from entering. This window type features a strong sash that closes tightly against the frame, which greatly contributes to low air leakage rates (*Window Types and Technologies*, n.d.). From a ventilation and daylight perspective, the full sash opening and unobstructed glass area of casement windows can provide fresh breezes into the tiny space and open clear views. This directly aligns with maintaining natural airflow and an enhanced connection to greenery and daylight. From more of a durability and usability point of view, the crank mechanism and insect screen compatibility make casement windows very functional in tight interiors. The trade-off of having these strong features is the need for higher hardware maintenance due to the operable crank and cleaning difficulties, depending on the placement. However, when considering the combination of airtight sealing, positive airflow and other advantages, the performance benefits of casement windows can fit the retreat's needs.

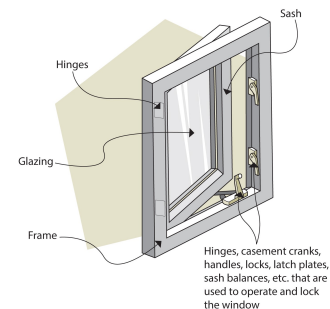


Figure 1: Casement window showing parts and hardware (Canada, 2014).

2.2 Awning Windows

Awning windows, on the other hand, are top-hinged and open outward, not from the side but from the bottom. These windows serve a strong advantage of being able to keep open during light rain, thus providing ventilation even during different weather conditions (Pizarro, 2024). Similar to Casement windows, this design is very energy efficient due to its tight seals, but this window has a smaller opening angle and is better suited for situations where the width of the structure exceeds the height (*Magic*, n.d.). From a ventilation perspective, the small opening angle has less airflow capacity compared to a fully opening sash. For a small interior area, having strong natural ventilation is a priority, as it can help direct odour and temperature. Additionally, since awning windows are typically shorter and wider, having visuals to external greenery and daylight will be limited compared to taller casement units. As a result, despite awning windows being a very versatile window in different weather conditions, their performance in ventilation and view area is less favourable, considering the priority of the problems.

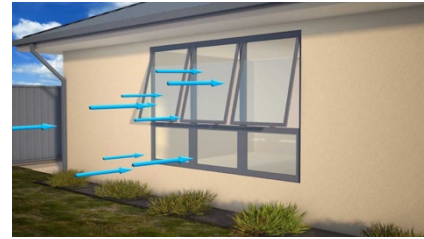


Figure 2: Awning window experience with exterior airflow (Breezway Louvre Windows, 2012).

2.3 Glazing and Frame Considerations

Regardless of the window type, the selection of glazing and frame materials is very important in determining thermal and overall performance. Windows and doors account for up to 25% of the total house heat loss; therefore, having proper and efficient windows with proper glazing and framing is critical (Canada, 2014). Selecting the proper frame material, glazing type, gas fills, and spacer technology all influence a window's energy performance (*Window Types and*

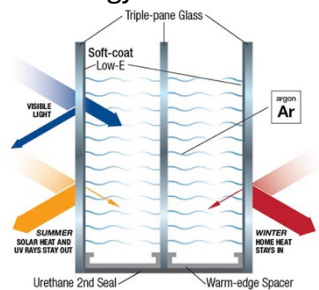


Figure 3: How argon gas filled insulation affects different factors to windows (*What Are the Benefits of Argon Gas in Windows?*, n.d.)

Technologies, n.d.). For a tiny retreat, a double-glazed insulated glazing unit incorporated with low-emissivity coating with argon fill is favourable to cost, weight and performance in comparison to a more expensive triple glazing. The spacers and insulated frames help further minimize the localized heat leakage and condensation issues. Due to the tiny space, this choice can maintain thermal stability while still considering high daylight, airflow, energy efficiency and more.

3. Conclusion and Recommendations

In conclusion, casement windows are the recommended choice as they provide better ventilation, maximize daylight and exposure to greenery, maintain good energy efficiency with low-E argon glazing and be able to function during both hot and cold climates. They are

compatible with insect screens and crank operation, which make them very suitable for small interiors. Awning windows may still serve as a secondary option for rain-protected ventilation, but casement windows provide better support to the retreat's needs of comfort, airflow and view. The final choice is a casement window with double-glazed, low-E, argon filled insulated glazing.

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